

How much water does a cooling scenario consume?

For researchers and planners comparing proposed data centers

INPUT

Capacity (MW)
Cooling type
Watershed + month

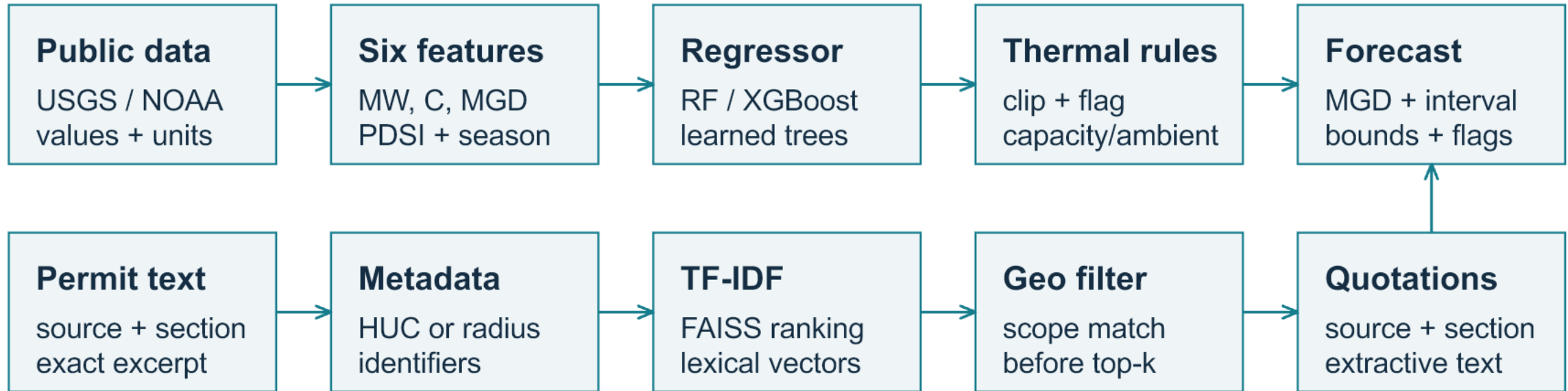
OUTPUT

Consumption (MGD) + bounds
Interval + quality warnings
Geographic permit quotations

Working local prototype | Shaun, Troy, Scott

Meaning lives in features, rules, vectors, and scope

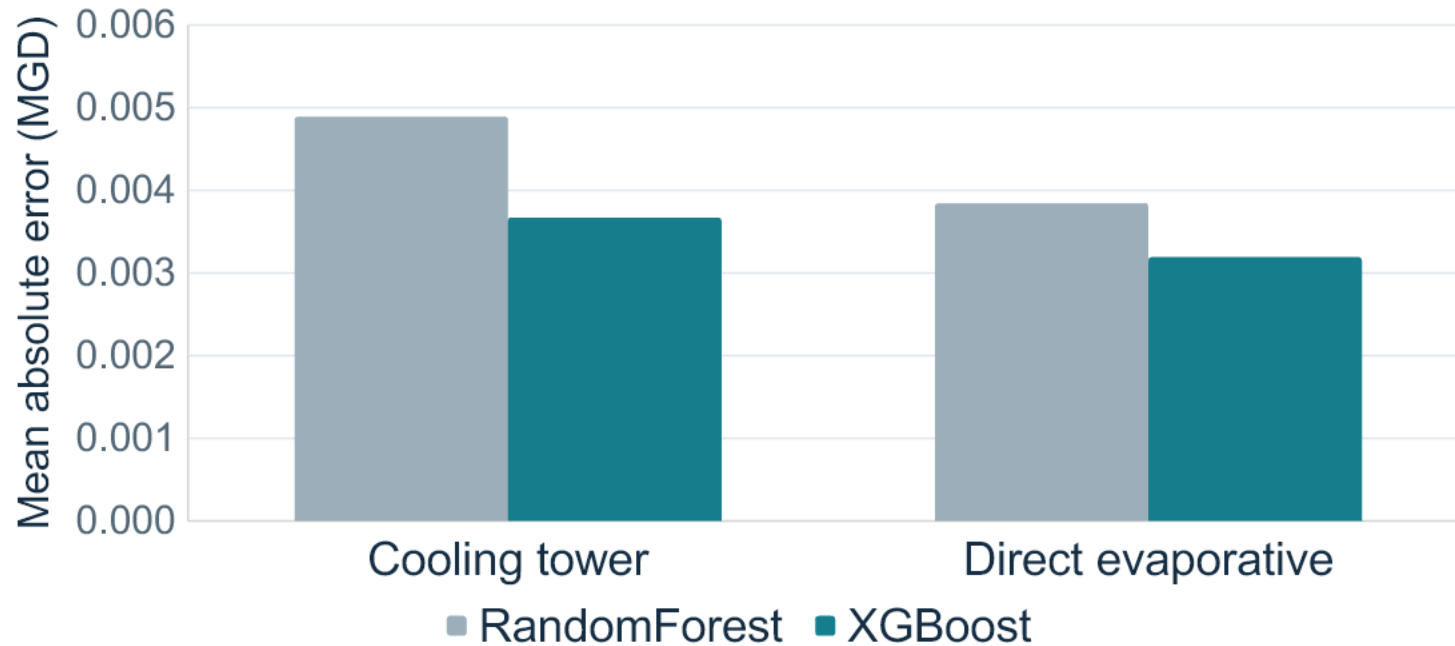
Request: MW, cooling type, HUC8, month; optional facility coordinates



Offline: synthetic consumption labels -> chronological train / calibration / test

Bounds flag injected errors; four geographic retrieval cases pass.

XGBoost reduced error on the synthetic benchmark



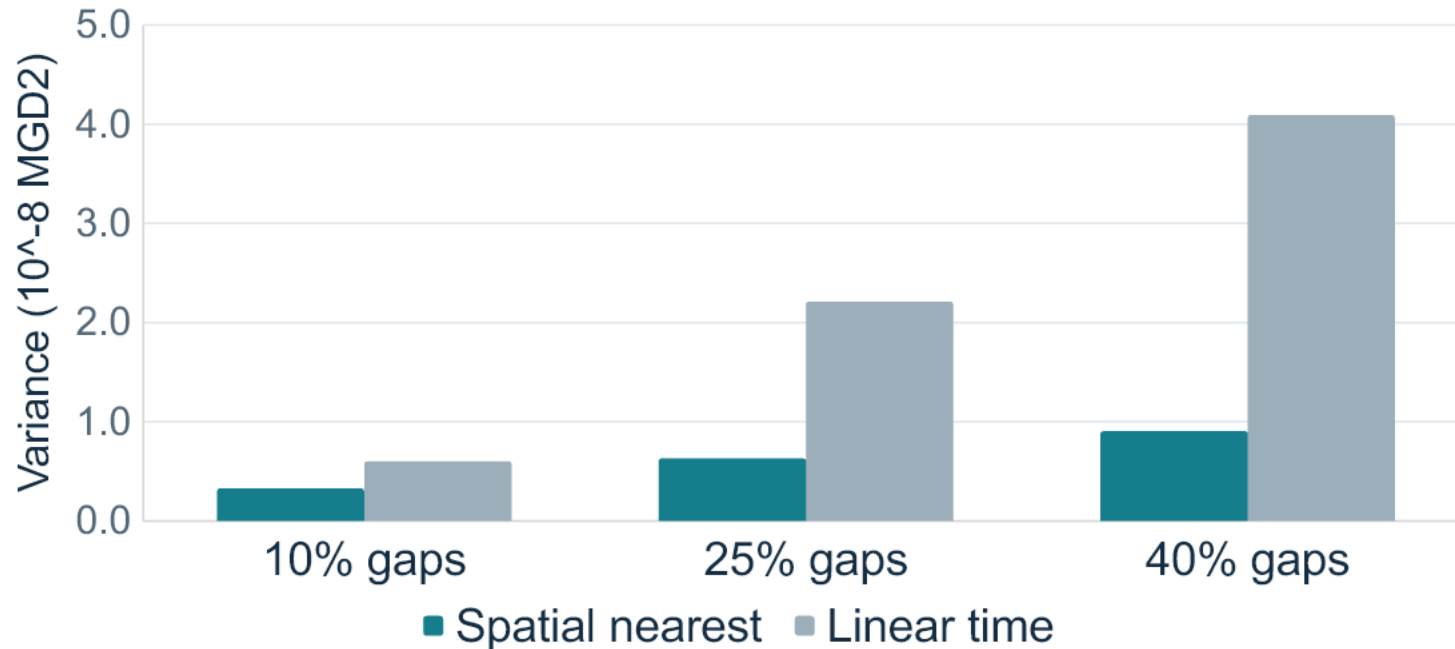
Adopt

Select by calibration
RMSE.
Keep RF as the
baseline.

**Measured facility
labels
remain the main gap.**

Chronological 60/20/20 dates | Dry air cooling: zero target, undefined R2

Correlated donors made drought gaps easy to fill



100% within tolerance

+/-0.05 MGD
50 repeats; 121 drought
dates

**Biased-donor
diagnostic:
spatial coverage 60.3%**

Modify: validate donors and shared outages; diagnostic uses a separate simple predictor.

High R2 did not establish 90% interval coverage



88.3% / 88.0%

Tower R2 = 0.998
Coverage misses the
observed 90% gate.

**Modify: separate tuning
and interval calibration.**

Finite samples; no significance test or independently validated coverage guarantee.

Next: test measured consumption at an unseen facility

Adopt

Tree baselines, independent rules, geography-aware quotations

Modify

Donor validation, shared outages, independent interval calibration

Reject / Defer

Reject R2-as-confidence; defer neural and generative LLM models

Can the constrained model generalize during drought with adequate coverage?

Shaun: code + calibration Troy: data + drought Scott: permits + evidence